PulmoTwin: Outpatient Respiratory Digital Twin

Slide 1: Title Slide

A 48-Hour Predictive Early Warning System for COPD & Asthma Flare-Ups

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- Affiliation: SRM University Delhi-NCR
- Team Name: <MY TEAM NAME>

- GitHub: https://github.com/MonalikaDoda/PulmoTwin
- Competition: Happiest Health Digital Twin Challenge 2026
- Stack: Python (FastAPI, XGBoost, SHAP) + React Dashboard (Vite, Three.js)
- License: MIT License | Research Prototype

The Crisis of Reactive Respiratory Care in Delhi NCR

Slide 2: Clinical Problem Statement

- Clinical Reality: Chronic respiratory diseases are a leading cause of emergency hospitalization in India. (Reference: Lancet Global Health / ICMR State-Level Disease Burden Initiative, 2018).
- Compound Urban Stressors: Delhi NCR experiences severe seasonal air quality episodes (CPCB NAQI > 300), extreme diurnal temperature swings, and mineral dust.
- The Critical Window: Airway inflammation begins 48–72 hours prior to acute decompensation (gradual SpO2 drift, mild tachycardia, nocturnal sleep reduction, and step count drop).
- The Gap: Current care is reactive—treating patients only after severe dyspnea forces an ER visit.

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PulmoTwin: 48-Hour Predictive Outpatient Digital Twin

Slide 3: Proposed Solution

- Multi-Modal Fusion: Fuses personal baseline vitals (SpO2, HR, RR, sleep, steps) with real modelled air quality (Open-Meteo CAMS) and weather (Open-Meteo ERA5).
- 48-Hour Early Warning: Identifies pre-flare physiological drift before acute hospitalization is required.
- Operational Feasibility: Averages 0.77 alerts per patient per month (~9.19 alerts/patient/year), grouping consecutive days into single multi-day episodes to prevent alert overload.
- Transparent Attribution: SHAP factor decomposition shows what shifted the estimate from the baseline.
- Counterfactual Sensitivity: Interactive What-If sliders show model response under hypothetical interventions.

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Real Modelled Weather/AQI + Synthetic Clinical Cohort

Slide 4: Data Sources & Provenance

- Synthetic Cohort (N=500 patients): Demographics, GOLD/GINA staging, smoking history, baseline adherence.
- Synthetic Telemetry (182,500 patient-days): 365 daily time-series with simulated pre-flare drift.
- Real Modelled Air Quality: Open-Meteo CAMS reanalysis mapped to Indian CPCB NAQI breakpoints (CC BY 4.0).
- Real Modelled Weather: Open-Meteo ECMWF ERA5-Land reanalysis (temperature, diurnal swing, dust, rain).
- Illustrative Pollen Index: Mathematical curve placed near seasons in Delhi pollen calendars; not measured data.
- Ethical Standard: Zero real patient data (no PHI). All patient data is synthetic.

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Empirical Benchmarks from model/comprehensive_metrics.json

Slide 5: Multi-Modal Ablation Study

Strict patient-level GroupShuffleSplit (400 train / 100 test patients; 36,500 test days; 0 leakage):

- M1 (Vitals Only, 17 feat): ROC-AUC 0.9982 | PR-AUC 0.9042 | Recall 94.09% | F1 0.8353
- M2 (Vitals + EHR, 27 feat): ROC-AUC 0.9981 | PR-AUC 0.9020 | Recall 93.82% | F1 0.8400
- M3 (Vitals + EHR + CAMS AQI, 35 feat): ROC-AUC 0.9982 | PR-AUC 0.9068 | Recall 93.82% | F1 0.8374
- M4 (Full Multi-Modal, 46 feat): ROC-AUC 0.9982 | PR-AUC 0.9078 | Recall 93.41% | F1 0.8358

Plain Finding: On this synthetic dataset, continuous vitals (M1) already yield PR-AUC 0.9042.
Adding environmental features (M4) provides virtually negligible gain (+0.0036 PR-AUC, +0.0005 F1),
showing that vital sign drift accounts for almost all predictive signal in the generator.

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Performance Under Warning Horizons and Missing Data

Slide 6: Lead-Time Warning & Dropout Robustness

Lead-Time Analysis (372 test flare events in test cohort):

- 48h Ahead Warning: 100.0% sensitivity on synthetic test labels
- 36h Ahead Warning: 100.0% sensitivity
- 24h Ahead Warning: 100.0% sensitivity
- Clinical Alert Workload: 0.77 alerts/pt/month (~9.19 alerts/year)

Wearable Data Random Dropout Robustness:

- 0% Missing: ROC-AUC 0.9982 | Recall 93.4% | Precision 75.6% | F1 0.8358
- 10% Missing: ROC-AUC 0.9963 | Recall 78.8% | Precision 75.3% | F1 0.7700
- 20% Missing: ROC-AUC 0.9925 | Recall 61.0% | Precision 78.7% | F1 0.6874
- 30% Missing: ROC-AUC 0.9861 | Recall 37.9% | Precision 78.5% | F1 0.5113

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Transparent SHAP Attribution & Sensitivity Modeling

Slide 7: Explainability & What-If Simulator

- SHAP TreeExplainer Attribution: Local log-odds converted to probability percentage points.
  Mathematically reconciled: Cohort Reference Prior (3.6%) + Bars = Displayed Risk%.
- 9 Factors: SpO2 vs baseline, resting HR, respiratory rate, sleep, steps, CAMS NAQI, dust, weather, adherence.
- Counterfactual What-If Simulator:
  - Allows interactive adjustment of indoor AQI reduction (0 to -150), adherence boost (+0 to +50%),
    and thermal swing dampening (0 to -8°C).
  - Shows model mathematical sensitivity on synthetic feature distributions;
    does NOT claim a proven clinical effect in real patients.

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Editorial-Clinical Hospital Monitoring UI

Slide 8: Interactive Clinical Dashboard

- Editorial Design: Warm off-white surface, deep teal accent, serif headings, light/glass cards.
- 3D Lung Digital Twin: Three.js breathing model with respiratory rate pacing and risk tinting.
- 5 Vital Tiles: SpO2, HR, RR, Sleep, Steps with 7-day sparklines and baseline deviation delta.
- 365-Day Live Simulation Engine: Play/Pause/Speed controls, date scrubber, and 'Jump to Next Alert'.
- Light/Glass Longitudinal Telemetry: Deep teal SpO2, crimson HR, amber CAMS NAQI, and steps bars.
- Delhi GIS Station Map: 5 ambient stations with patient pinned and micro-climate readings.
- Error Boundary: Robust fault isolation preventing blank screen failures.

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Transparent Statement of Limitations

Slide 9: Scientific & Clinical Limitations

1. Synthetic Patient Cohort: All static EHR and continuous wearable records are synthetic.
   No real patient data was utilized.
2. Label Rule Recovery: Flare-up labels were generated according to our own physiological drift simulation
   rules; high performance reflects accurate recovery of these rules on synthetic data.
3. No Real Patient Clinical Evidence: There is no prospective or retrospective evidence showing this model
   operates safely or effectively in clinical practice.
4. Modelled Atmospheric Data: Air quality and weather are from Open-Meteo reanalysis models (CAMS and ERA5),
   not physical hardware monitoring stations.
5. Illustrative Pollen Index: The pollen formula is an illustrative seasonal mathematical curve, not measured data.
6. Decision-Support Only: Research prototype for decision-support; not a diagnostic tool.

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Project Deliverables & Repository Information

Slide 10: Summary & Submission Deliverables

- Team: Monalika Doda, SRM University Delhi-NCR, <MY TEAM NAME>, solo.
- Repository: https://github.com/MonalikaDoda/PulmoTwin
- Stack: Python 3.10+ (FastAPI, XGBoost, SHAP, Pandas) + React 18 (Vite, Three.js).
- Deliverables in docs/:
  - docs/PulmoTwin_Architecture_Diagram.pdf (Vector pipeline)
  - docs/PulmoTwin_Presentation_Deck.pdf (Executive pitch deck)
  - docs/architecture.md and docs/deck_outline.md
- License: MIT License (Copyright (c) 2026 Monalika Doda)
- Development: Built with AI coding assistance (Antigravity).
- Public Access: Both backend (port 8000) and frontend (port 3000) run locally with zero external API keys.